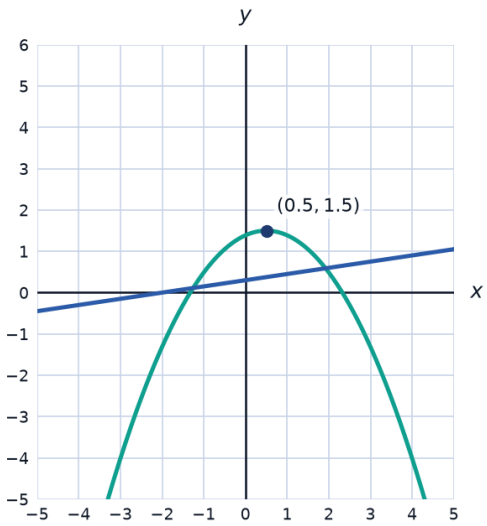
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int x^4\,dx$.

2 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \sqrt{x}\,dx$.



3 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \frac{1}{x^3}\,dx$.

4 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (4x^3 - 6x + 5)\,dx$.

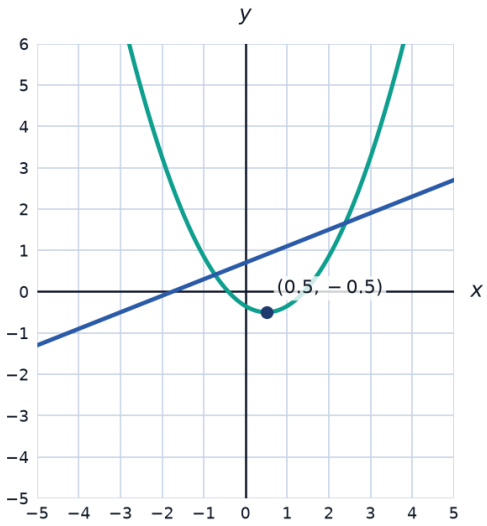
5 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (2x + 7)\,dx$.

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6 [Foundational] • Target: Antiderivatives of polynomials

Find $\int (x^2 - 4x + 3)\,dx$.



7 [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \cos x\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \sin x\,dx$.

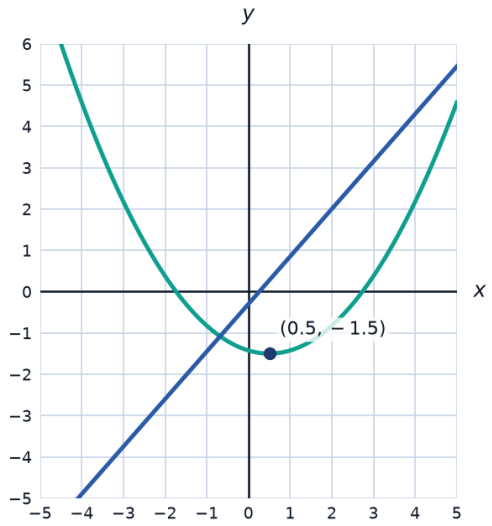
9 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int e^x\,dx$.

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10 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^2 3x^2\,dx$.



11 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_1^3 2x\,dx$.

12 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_{-1}^2 3x^2\,dx$.

13 [Intermediate] • Target: Definite integrals with polynomial integrands

Evaluate $\int_0^2 (x^2+1)\,dx$.

14 [Intermediate] • Target: Definite integrals with polynomial integrands

Evaluate $\int_1^3 (2x^2-3x+1)\,dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

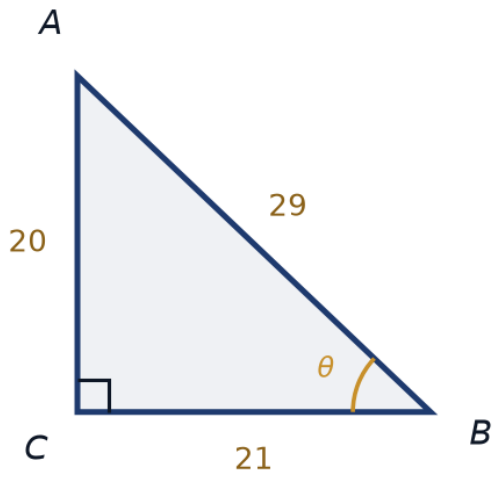
15 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\int_{-2}^1 (x^2-x)\,dx$.

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16 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\int_0^{\pi/2} \cos x \, dx$.



17 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

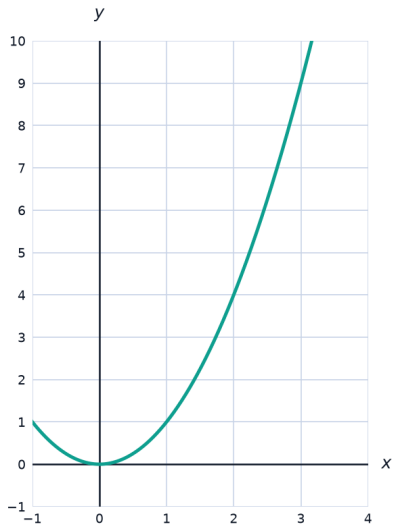
Evaluate $\int_0^{\pi} \sin x \, dx$.

18 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_0^{1} e^x \, dx$.

19 [Challenging] • Target: Area under a curve • Cross-Lesson: 1.01 Algebra & Equations

Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.



20 [Challenging] • Target: Area under a curve • Cross-Lesson: 2.01 Functions & Transformations

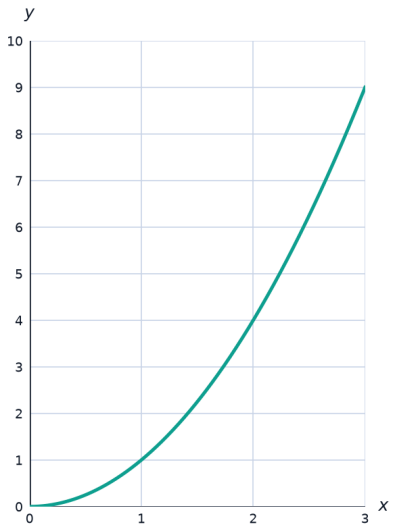
Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Area under a curve**

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.



a. 9

b. 18

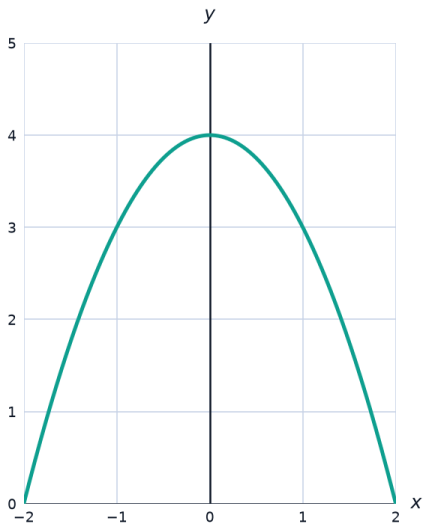
c. 3

d. 27

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22 [Foundational] • Target: Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.



a. $\frac{16}{3}$

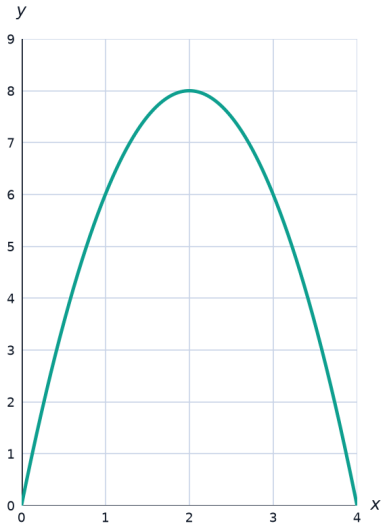
b. 32

c. 16

d. $\frac{32}{3}$

23 [Foundational] • Target: Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).



a. $\frac{64}{3}$

b. 16

c. 32

d. 64

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24 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \sqrt[3]{x}\, dx$.

a. $\frac{3}{2}x^{2/3} + C$

b. $\frac{4}{3}x^{4/3} + C$

c. $\frac{3}{4}x^{4/3} + C$

d. $\frac{1}{3}x^{-2/3} + C$

25 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int 7\, dx$.

a. C

b. $7x + C$

c. $7 + C$

d. $\frac{7x^2}{2} + C$

26 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \frac{1}{\sqrt{x}}\, dx$.

a. $2\sqrt{x} + C$

b. $\frac{1}{2}\sqrt{x} + C$

c. $-2\sqrt{x} + C$

d. $\sqrt{x} + C$

27 [Foundational] • Target: Antiderivatives of polynomials

Find $\int (3x^3 - 2x)\, dx$.

a. $3x^4 - x^2 + C$

b. $\frac{3x^4}{4} - x^2 + C$

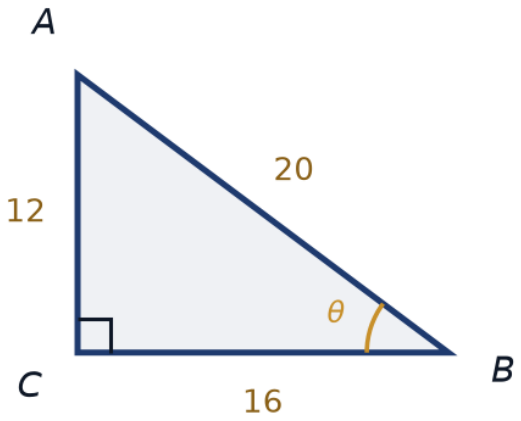
c. $\frac{3x^4}{4} - 2x^2 + C$

d. $\frac{x^4}{4} - x^2 + C$

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28 [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int 2\cos x\,dx$.



a. $2\sin x + C$

b. $-2\sin x + C$

c. $2\cos x + C$

d. $\sin(2x) + C$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int 3\sin x\,dx$.

a. $3\sin x + C$

b. $-3\sin x + C$

c. $3\cos x + C$

d. $-3\cos x + C$

30 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int (e^x + 2)\,dx$.

a. $e^x + 2 + C$

b. $e^x + 2x + C$

c. $xe^x + 2x + C$

d. $e^x + C$

31 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int (\sin x + \cos x)\,dx$.

a. $-\cos x - \sin x + C$

b. $\cos x + \sin x + C$

c. $\cos x - \sin x + C$

d. $-\cos x + \sin x + C$

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32 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_2^5 3\,dx$.

a. 3

b. 9

c. 6

d. 15

33 [Intermediate] • Target: Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi/2} \sin x\,dx$.

a. $\frac{\pi}{2}$

b. -1

c. 0

d. 1

34 [Intermediate] • Target: Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi} \cos x\,dx$.

a. π

b. -2

c. 0

d. 2

35 [Intermediate] • Target: Signed area and regions below the axis

Evaluate $\int_0^4 (x-4)\,dx$ and state whether the region is above or below the x-axis.

a. 8; above

b. -8; above

c. 8; below

d. -8; below

36 [Intermediate] • Target: Properties of definite integrals

Given $\int_1^5 f(x)\,dx = 20$ and $\int_1^5 3\,dx = 12$, find $\int_1^5 [f(x) - 3]\,dx$.

a. -8

b. 8

c. 32

d. 17

37 [Intermediate] • Target: Properties of definite integrals

Given $\int_0^4 f(x)\,dx = 9$ and $\int_0^4 g(x)\,dx = 5$, find $\int_0^4 [f(x) + g(x)]\,dx$.

a. $\frac{9}{5}$

b. 45

c. 4

d. 14

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Finding a function from its derivative • Cross-Lesson: 9.01 Statistics & Probability

A function satisfies $f'(x) = 2x + 7$ and $f(-1) = 4$. Find $f(x)$.

a. $x^2 + 7x + 10$

b. $2x^2 + 7x + 10$

c. $x^2 + 7x - 10$

d. $x^2 + 7x + 4$

39 [Challenging] • Target: Average value of a function • Cross-Lesson: 1.01 Algebra & Equations

Find the average value of $f(x) = x^2 - 1$ on $[1, 3]$.

a. $\frac{8}{3}$

b. $\frac{10}{3}$

c. $\frac{20}{3}$

d. 6

40 [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 2.01 Functions & Transformations

A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \le t \le 4$. Find the particle's displacement over this interval.

a. 64

b. $\frac{64}{3}$

c. 32

d. $\frac{192}{3}$

41 [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 3.01 Sequences & Series

Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \le t \le 10$. Find the total volume of water that flows in during this time.

a. 300

b. 150

c. 450

d. 100

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42 [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 4.01 Trigonometry

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

a. 528

b. 656

c. 450

d. 400

43 [Challenging] • Target: Integration & Applications • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find $\int -x^4\,dx$.

a. $-\frac{1}{5}x^5 + C$

b. $-\frac{1}{4}x^4 + C$

c. $-x^5 + C$

d. $-\frac{1}{6}x^5 + C$

44 [Challenging] • Target: Integration & Applications • Cross-Lesson: 6.01 Limits & Derivatives

Find $\int -3x^4\,dx$.

a. $-3x^5 + C$

b. $-\frac{3}{5}x^5 + C$

c. $-\frac{1}{2}x^5 + C$

d. $-\frac{3}{4}x^4 + C$

45 [Challenging] • Target: Integration & Applications • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\int (-2x + 5)\,dx$.

a. $-2x + 5 + C$

b. $-x^2 + 7x + C$

c. $-x^2 + 5x + C$

d. $-x^2 - 5x + C$

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